Six seeds of d128 still miss the free line, so the minimum holds at 2.50M

stage N -- KB on fresh seeds 3/4/5, pooled with stage K's 0/1/2

The pooled call (N2)

AFTER -- 0.761 at 1,145,658 trainable, 6/6 crossing, median 0.761 is +0.14 s* past the free line 0.740. The minimum does NOT move; it stays 0.580 at 2,504,890, unchanged by this stage.

The previous best, same eval

BEFORE -- 0.580 at 2,504,890 trainable (`no_mlp L=16 d192`, J2, finding 53), measured from disk on this same pooled eval as 0.580 over seeds 0.580/0.557/0.752, 3/3 crossing.

Why the rule was fixed first

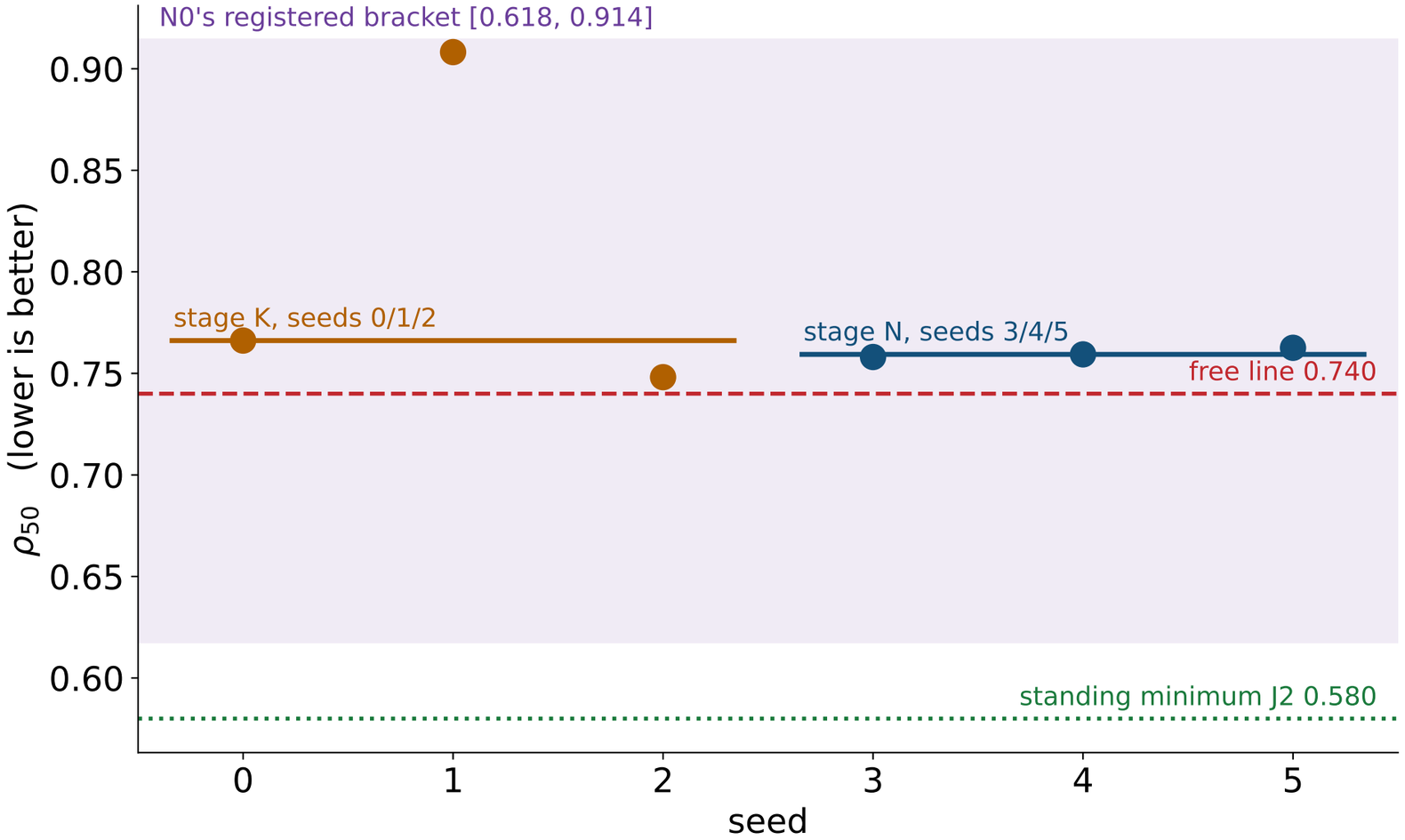
N2's pooling rule was fixed at registration -- 6 of 6 seeds crossing AND the six-seed median at or below 0.740 -- because choosing how to pool after seeing the fresh numbers is how a narrow miss becomes a hit. Stage K's three seeds gave 0.766, a miss of +0.18 s*.

What size this is about

This is an N=48 claim. Finding 55 measured that the N=48 minimum does not transfer to N=96, so nothing on this page says anything about N=96, and the two sizes' rho_50 are never pooled.

Both the level and the crossing replicate on fresh seeds

one rung, two independent seed triples, identical in every other flag




```
N0 -- did the LEVEL replicate?
  registered  fresh median in
              [0.618, 0.914]
  measured    0.759
  -> HELD

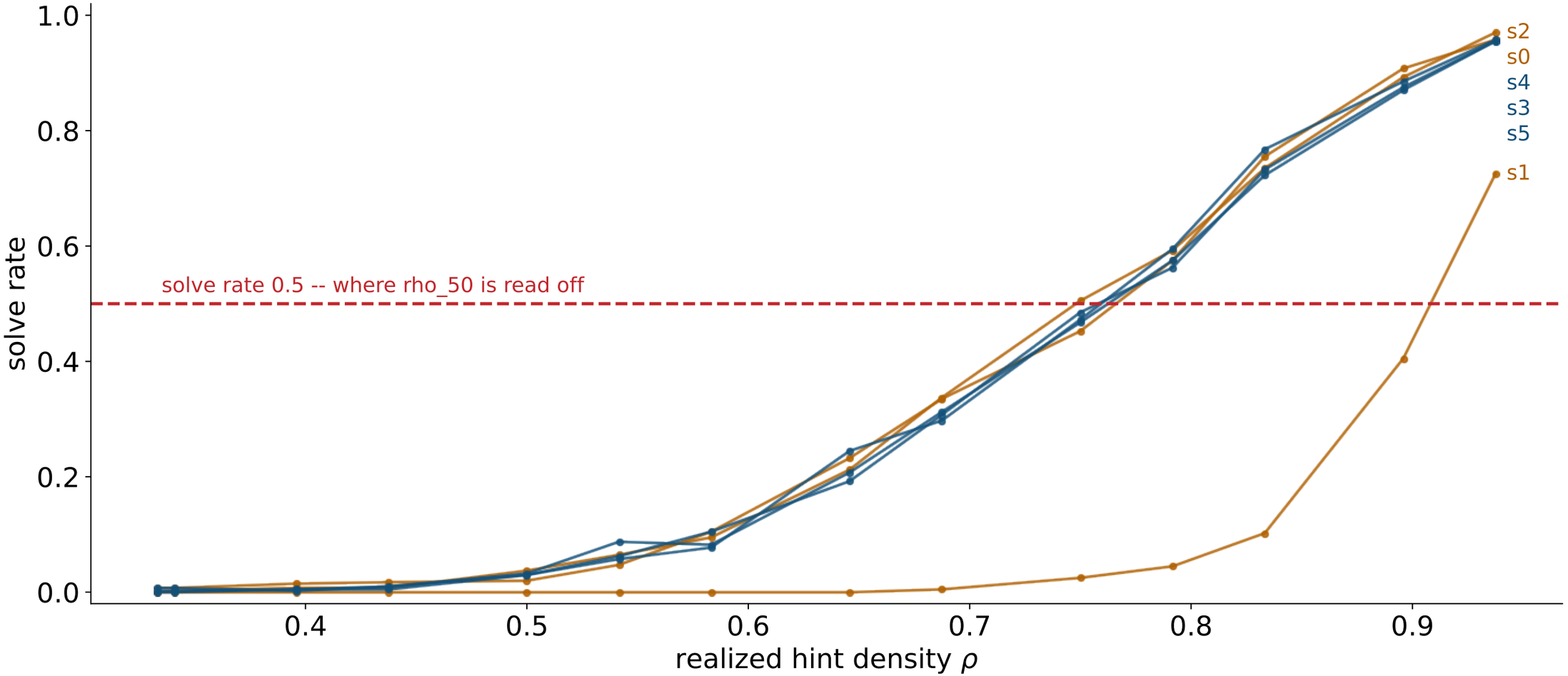
N1 -- did the CROSSING replicate?
  registered  3 of 3 fresh seeds
  measured    3 of 3
  -> HELD

N1 and N3 are read BEFORE N2:
a level from a censoring rung
is a bound, not a measurement.
```

commit 43e013b · every seed crossed, so no marker is a censor · horizontal bars are each triple's bound-aware median · a censor is a result, not a missing number

1 seed that never left the plateau carries the whole spread

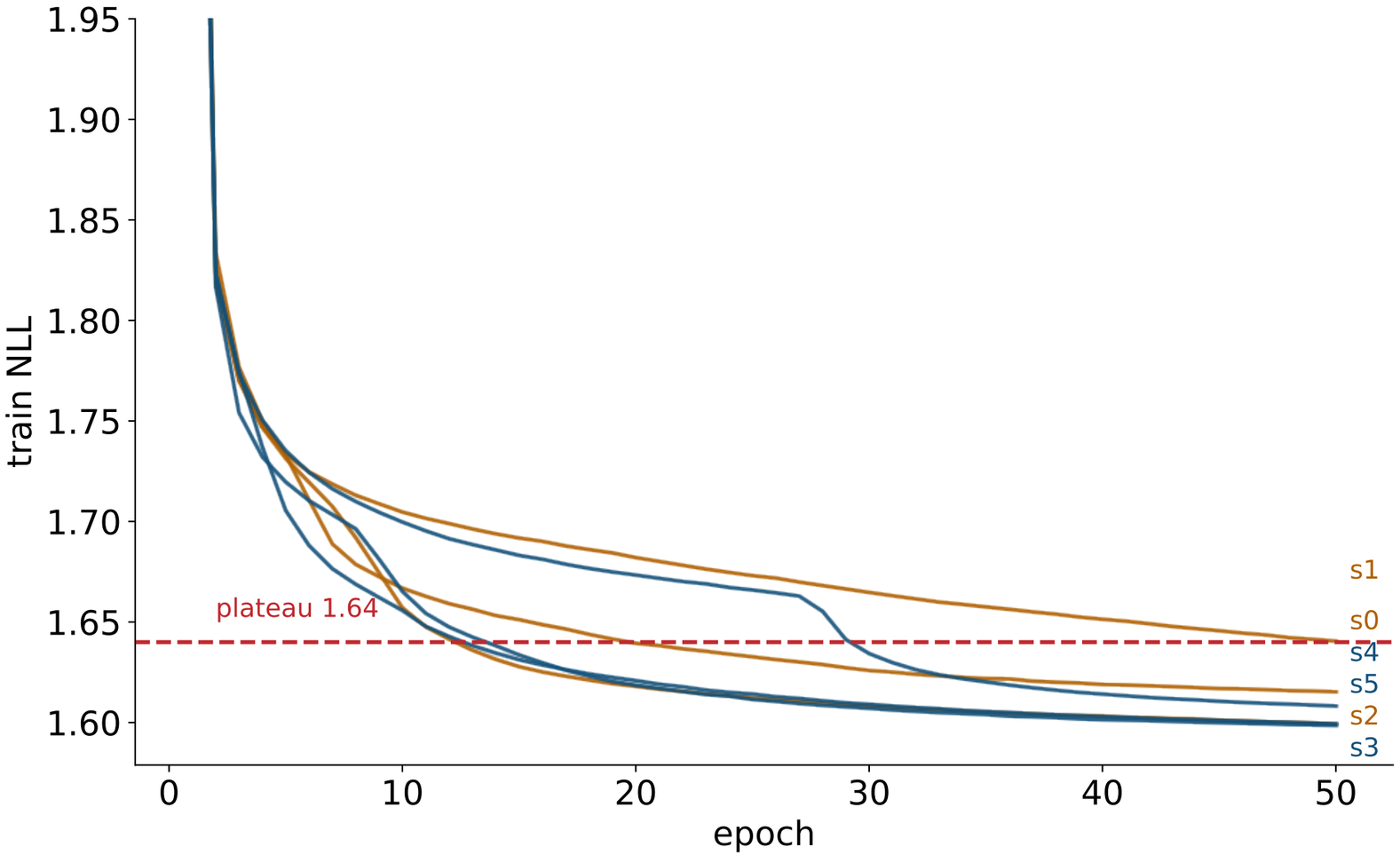
solve rate against realized hint density -- the curve rho_50 is read off



commit 43e013b · orange = stage K's seeds 0/1/2, blue = stage N's 3/4/5 · 400 instances per bin · grid floor 0.33, top 0.95 · rho_50 censors when no bin clears 0.5

5 of 6 seeds train out of the plateau, so the level is capacity

training loss against the N=48 plateau -- capacity or trainability



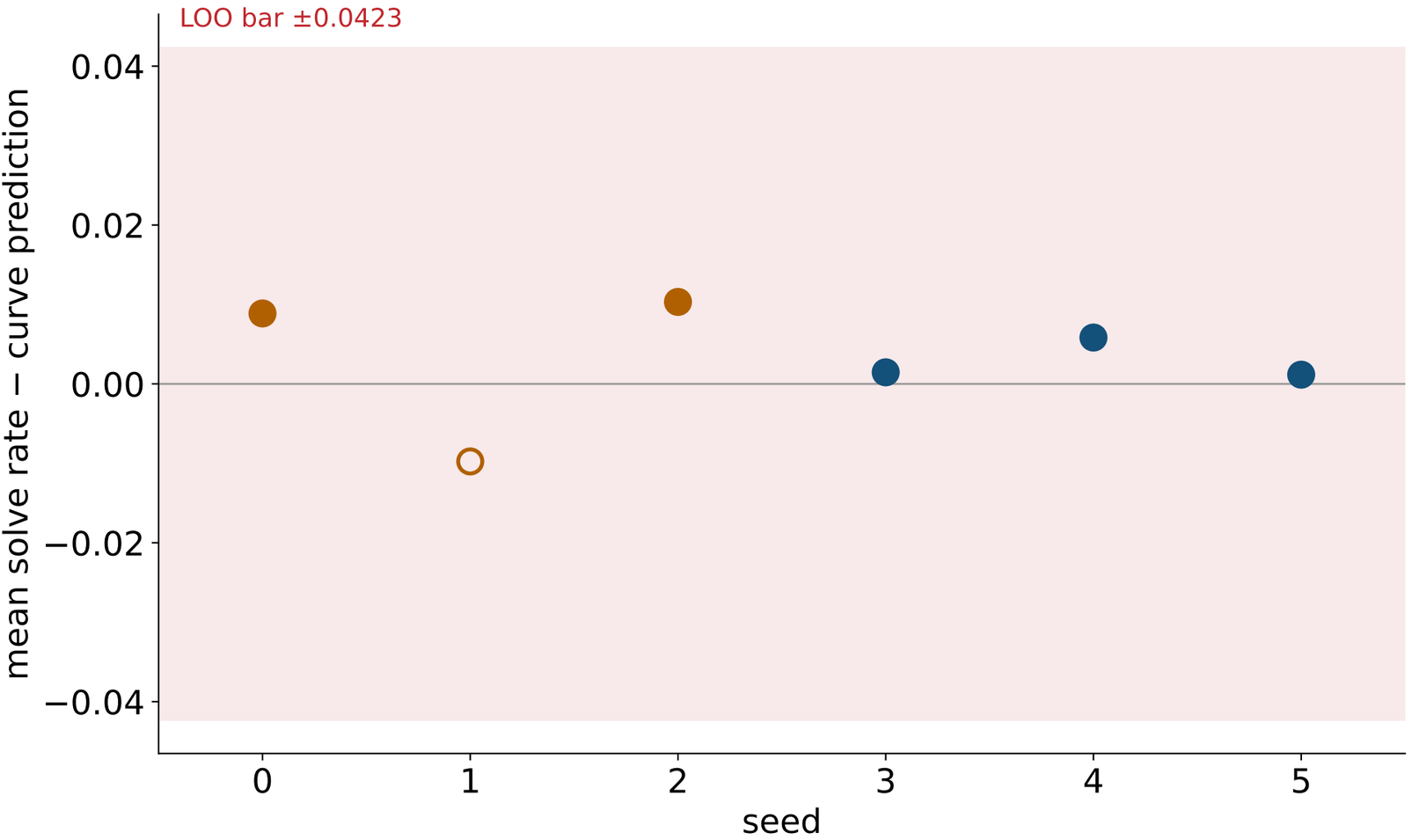

```
N3 -- the plateau
  registered  >= 4 of 6 escape
  stage K     2 of 3
  stage N     3 of 3
  pooled      5 of 6
  -> HELD

final train NLL
  s0  1.615
  s1  1.641  never escaped
  s2  1.599
  s3  1.599
  s4  1.608
  s5  1.600
```

commit 43e013b · the plateau 1.64 is finding 44, an N=48 number · orange = stage K, blue = stage N · y-axis cropped at 1.95; epoch 1 starts near 2.4 · every checkpoint is the final epoch, no best-val selection · an optimization-limited rung outranks N2

All 5 discriminating seeds sit on the curve, worst miss 0.0103

distance from the depth ladder's accuracy-to-solve curve



N4 -- on the curve?
  bar ±0.0423

seed	acc	residual	
s0	0.4817	+0.0089	
s1	0.4701	-0.0098	wide
s2	0.4831	+0.0103	
s3	0.4829	+0.0015	
s4	0.4827	+0.0058	
s5	0.4833	+0.0012	

Accuracy is read BEFORE rho_50 because it never censors, and a rung that falls OFF this curve is the interesting one. Hollow points are not discriminating: the prediction is on the floor, or its bracketing reference cells are further apart than the bar, so the residual is drawn but claims nothing.

commit 43e013b · orange = stage K, blue = stage N · curve fitted by scripts/acc_vs_solve.py over the depth ladder only
· N4's held-out sufficiency check runs separately

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-57	`N0 N1 ... N47`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	`BEGIN_HINTS N2 COLOR_0 N5 COLOR_1 N6 COLOR_0 N10 COLOR_0 N12 COLOR_3 N13 COLOR_2 ... (39 tokens)`
GRAPH	`BEGIN_GRAPH N0 N23 N15 N26 N4 N31 N1 N29 N5 N7 N6 N22 ... (289 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N1 COLOR_2 N9 COLOR_1 N20 COLOR_3 N23 COLOR_0 N24 COLOR_3 N33 ... (73 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 29 deductions,
58 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000004, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

19 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N2=COLOR_0 N5=COLOR_1 N6=COLOR_0 N10=COLOR_0 N12=COLOR_3 N13=COLOR_2 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N1 COLOR_2 N9 COLOR_1 N20 COLOR_3 N23 COLOR_0 ...
  10 deduced
BEGIN_ROUND        N0 COLOR_1 N3 COLOR_2 N11 COLOR_1 N25 COLOR_1 ...
  6 deduced
BEGIN_ROUND        N19 COLOR_0
  1 deduced
BEGIN_ROUND        N34 COLOR_3
  1 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N1, neighbours N29 N5 N7 N6 N22

their colours known before this round:

```
[0, 1, 3]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N7 N22
```

only colour left: 2

```
so the trace emits  N1 COLOR_2
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. Stage N changes nothing about it and nothing about the architecture either -- only the seed triple moves, from 0/1/2 to 3/4/5.

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

The minimum holds, but a rung below d128 is still resolvable

what stage N settles, and what it leaves open

The negative result, stated as one

6 seeds of the smallest crossing rung still do not make it free: the six-seed median is +0.14 s* past 0.740. `no_mlp L=16 d192` keeps the minimum at 2,504,890 trainable, 0.580, unchanged by this stage.

What it does settle

Stage K's KB row is no longer a three-seed number. Its level and its crossing now rest on 6 seeds, which is what K1's own registration asked for before the d128 result is quoted outside the scorer.

Where the next cut is

d128 -> d192 moves rho_50 by 0.181. The raw six-seed spread is 0.160, but all of it is the 1 seed that never escaped the plateau; the 5 that did span only 0.018. Conditioned on N3's criterion -- pre-registered, not chosen after the fact -- the width effect is about 10x the seed spread, so a rung BELOW d128 is resolvable and is the next cut. Three seeds are not enough to find the stuck one, which is the real cost.